Calculus I Assignment 1

1. Find the numbers that satisfy each inequality, and sketch the solution on the real line.

   (a) $|x - 3| \leq 4$

   (b) $|x + 50| \leq 2$

   (c) $1 < |y - 7|$

   (d) $|3 - x| < 4$

2. Which inequalities are true for all numbers a and b satisfying $0 < a < b < 1$?

   (a) $ab > 1$

   (b) $\frac{1}{a} < \frac{1}{b}$

   (c) $\frac{1}{b} > 1$

   (d) $a + b < 1$

   (e) $a^2 < 1$

   (f) $a^2 + b^2 < 1$

   (g) $a^2 + b^2 > 1$

   (h) $\frac{1}{a} > b$

3. What would you choose for m in $|\sqrt{3} - 1.7| < 10^{-m}$, and why?

4. Find the least upper bound and the greatest lower bound of each of the following sets. Or if it is not possible, explain why.

   (a) the interval $(8, 10)$.

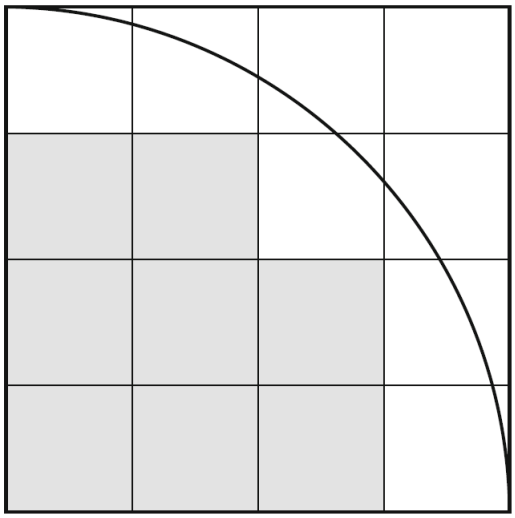
   (b) the interval $(8, 10]$.

   (c) the set of all non-positive integers.

   (d) the set of four numbers $\{\frac{30}{279}, \frac{29}{263}, \frac{59}{525}, \frac{1}{9}\}$.

   (e) the set $\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$

5. Take the unit square, and by connecting the midpoints of opposite sides, divide it into $2^2 = 4$ subsquares, each of side 2^{-1}. Repeat this division for each subsquare, obtaining $2^4 = 16$ squares whose sides have length 2^{-2}. Continue this process so that after n steps, there are 2^{2n} squares, each having sides of length 2^{-n}. See the following figure.



With the lower left corner as center and radius 1, inscribe a unit quarter circle into the square. Denote by a_n the total area of those squares that at the nth step of the process, lie entirely inside the quarter circle. For example, $a_1 = 0, a_2 = \frac{1}{4}, a_3 = \frac{1}{2}$.

(a) Is the set S of numbers $a_1, a_2, a_3, \ldots, a_n, \ldots$ bounded above? If so, find an upper bound.

(b) Does S have a least upper bound? If so, what number do you think the least upper bound is?

6. Define $x_n = \frac{n}{n+1}$:

$$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \cdots, \frac{n}{n+1}, \cdots$$

(a) Plot the points of this sequence on the real line (only the first eight terms are required);

(b) Prove $\lim_{n\to\infty} x_n = 1$;

(c) For the given ϵ in the table below, fill in the corresponding N:

ϵ	0.01	0.009	0.008	0.007	0.006	0.005	0.004	0.003	0.002	0.001
N										

7. Write out the first ten terms of the following sequences, and indicate whether they have a limit:

(a) $x_n = \frac{1+(-1)^n}{3} \quad (n = 1, 2, 3, \cdots)$;

(b) $x_n = (-1)^n \frac{n}{(n+1)} \quad (n = 1, 2, 3, \cdots)$.

8. Prove that:

(a) $\lim_{n\to\infty} \frac{2n}{n^3+1} = 0$;

(b) $\lim_{n\to\infty} \frac{3n+5}{2n-3} = \frac{3}{2}$.

9. Show that if $|q|<1$, then

$$\lim_{n\to\infty} q^n = 0.$$

Bonus Questions

- Bonus questions, while typically more challenging, are not compulsory.
- The maximum achievable score, including both regular and bonus questions, is capped at 100 points. For instance, if you score 96 points on the regular questions and an additional 10 points on bonus questions, your total score will be adjusted to the maximum limit of 100 points, not 106.

1. Suppose $a > 0$. Prove that $\lim_{n\to\infty} \sqrt[n]{a} = 1$.
2. Prove that $\lim_{n\to\infty} \frac{2n}{n!} = 0$.
3. Prove that $\lim_{n\to\infty} \sqrt[n]{n} = 1$.